Alan Pysnack

apysnack@gmail.com — 240-543-2916 — linkedin.com/in/apysnack — github.com/apysnack

Research Interests

Artificial Intelligence, Human-Computer Interaction, Computational Social Science, and Software Systems

Education

University of Maryland, Baltimore County (UMBC) 2026–Present
M.S. in Computer Science *Baltimore, MD*

University of South Alabama December 2021
B.S. in Computer Science, summa cum laude *Mobile, AL*
GPA: 4.0/4.0

Research Experience

University of Maryland, Baltimore County Fall 2026
Graduate Research Project — CMSC 601: Research Skills *Baltimore, MD*

- Designing a study investigating whether an AI-based social game can facilitate conversation and social connection between strangers.
- Building a prototype application to collect participant survey data and produce personalized, AI-generated visual artifacts for use during the experiment.
- Developing an experimental design to evaluate participants' enjoyment, perceived social connection, and willingness to interact again.

University of South Alabama Fall 2021
Undergraduate Research Assistant *Mobile, AL*

- Trained and tested supervised machine-learning classification algorithms to predict signal peptides in eukaryotic and archaeal protein sequences using Weka.
- Evaluated classification results using Matthews correlation coefficient (MCC).
- Prepared datasets and compared the performance of different classification algorithms.

Research & Technical Projects

TPM 2.0 PCR Fragility — Senior Capstone Project 2021
University of South Alabama

- Investigated PCR fragility in TPM 2.0 systems, where changes to measured platform state, such as firmware updates, can invalidate policies protecting sealed encryption keys.
- Studied the TPM 2.0 Software Stack and Kmyth, an open-source TPM security project maintained by the National Security Agency; researched and demonstrated two approaches for accommodating changes in platform state: TPM 2.0 Policy OR and Policy Authorization using the TPM Software Stack System API (SAPI).
- Implemented the selected Policy OR approach in Kmyth's existing C codebase, extending TPM sealing and unsealing functionality to support multiple PCR policy states.
- Added functionality for obtaining expected PolicyPCR values and incorporating alternative policy digests into the Kmyth sealing workflow.
- Contributed the Policy OR implementation to Kmyth, an open-source TPM security project maintained by the National Security Agency.

Publications

Pysnack, A. "Using AI-Generated Images to Facilitate Social Connection Between Strangers." *Imaginary*, 2026.

Presentations

Pysnack, A. "Addressing PCR Fragility in TPM 2.0 Systems." Senior Capstone Presentation, University of South Alabama, 2021.

Professional Experience

Assurant, Inc. February 2022–January 2026
Associate Software Engineer

- Developed and maintained production web and mobile applications using Ruby on Rails, React, React Native, and TypeScript.
- Designed and implemented RESTful and GraphQL APIs using Rails, Grape, and GraphQL.
- Optimized pagination logic and client-side caching, achieving an approximately 2,000-fold reduction in API requests for a high-volume workflow.
- Wrote unit, integration, and end-to-end tests using RSpec, Jest, and Playwright.
- Built internal tools for API simulation and logging to improve cross-team debugging and development workflows.
- Participated in code reviews, feature planning, technical design, and architecture discussions.

Genzeon Technology Solutions June 2021–September 2021
OCR Validation Intern

- Collaborated with product teams to evaluate OCR systems for accurate extraction of medical billing data.
- Developed Python tools to transform PDF documents into structured JSON using Azure Form Recognizer.
- Automated PDF ingestion and processing in Azure using a scheduled monitoring script.

Service

No formal academic service experience.

Technical Skills

Languages: JavaScript, TypeScript, Ruby, Python, C, C++, Java, SQL

Frameworks & Libraries: Ruby on Rails, React, React Native, Redux, Flask, GraphQL

Tools & Platforms: AWS, Azure, Docker, Git, GitHub Actions, Heroku